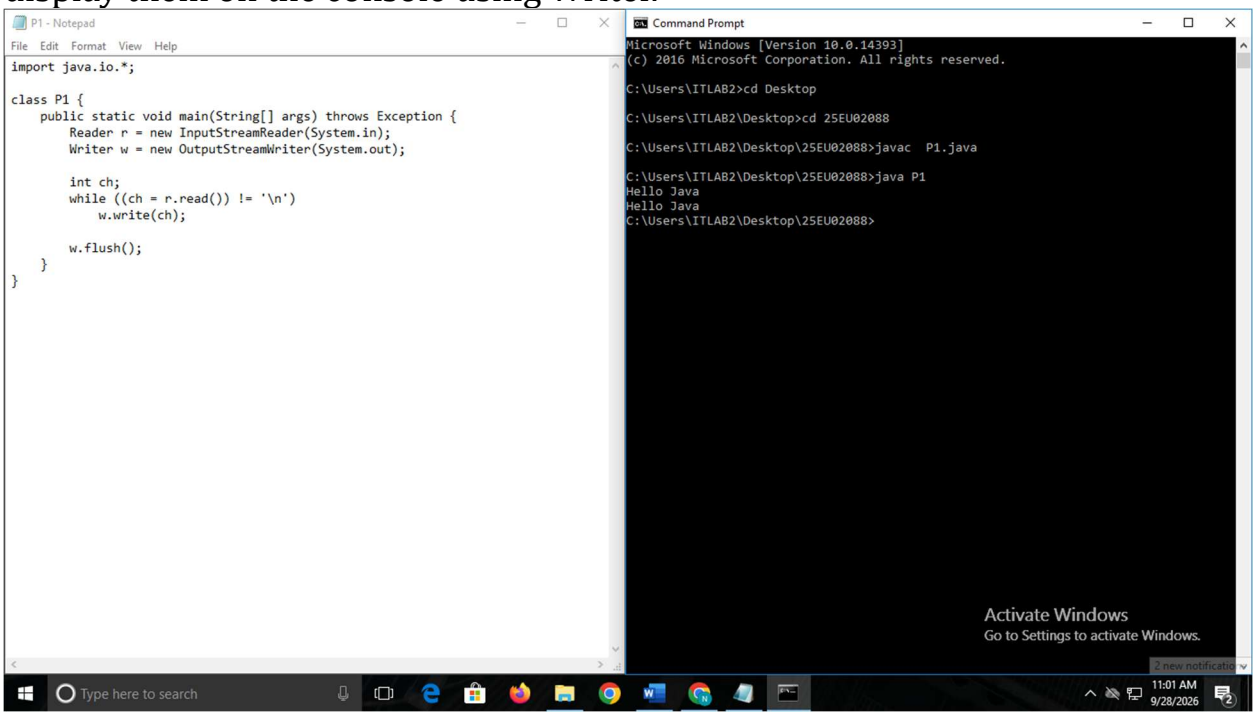
The Character Streams and Multithreaded Programming

1.Write a Java program to read characters from the keyboard using Reader and display them on the console using Writer.



P1 - Notepad

File Edit Format View Help

```
import java.io.*;

class P1 {
    public static void main(String[] args) throws Exception {
        Reader r = new InputStreamReader(System.in);
        Writer w = new OutputStreamWriter(System.out);

        int ch;
        while ((ch = r.read()) != '\n')
            w.write(ch);

        w.flush();
    }
}
```

Command Prompt

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2>cd Desktop

C:\Users\ITLAB2\Desktop>cd 25EU02088

C:\Users\ITLAB2\Desktop\25EU02088>javac  P1.java

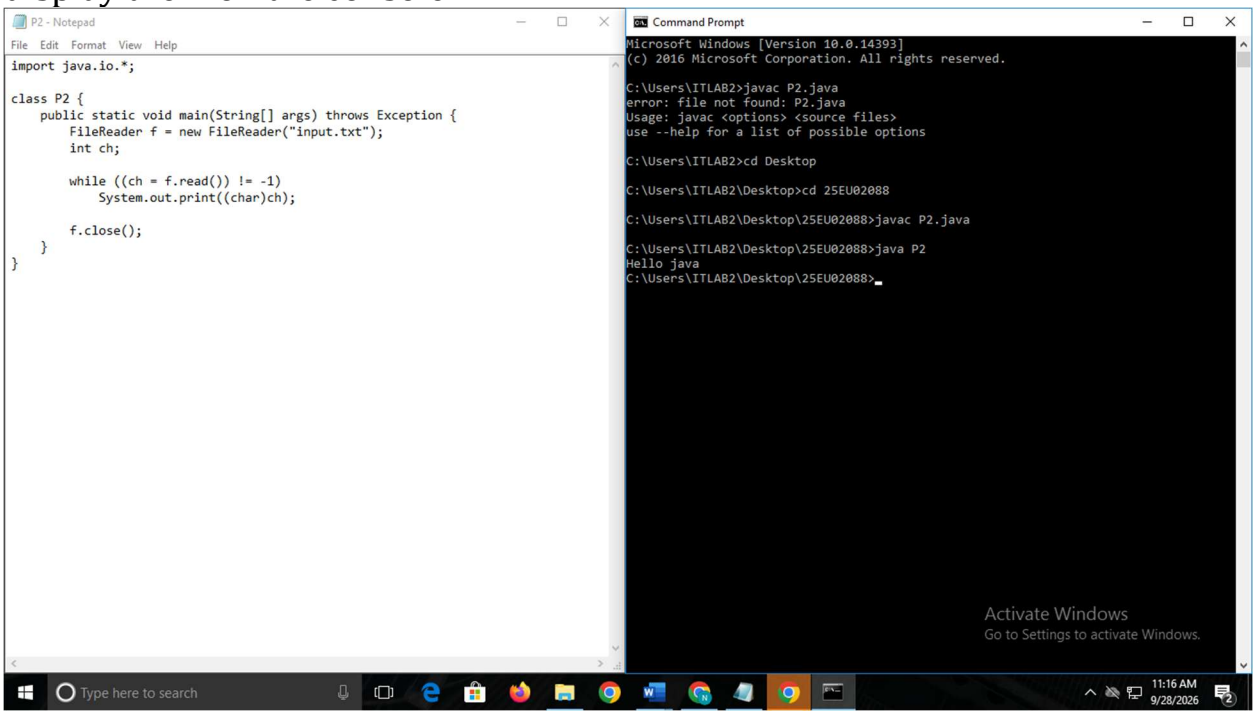
C:\Users\ITLAB2\Desktop\25EU02088>java P1
Hello Java
Hello Java
C:\Users\ITLAB2\Desktop\25EU02088>
```

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Go to Settings to activate Windows.

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2. Write a Java program to read the contents of a text file using FileReader and display them on the console.



P2 - Notepad

File Edit Format View Help

```
import java.io.*;

class P2 {
    public static void main(String[] args) throws Exception {
        FileReader f = new FileReader("input.txt");
        int ch;

        while ((ch = f.read()) != -1)
            System.out.print((char)ch);

        f.close();
    }
}
```

Command Prompt

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2>javac P2.java
error: file not found: P2.java
Usage: javac <options> <source files>
use --help for a list of possible options

C:\Users\ITLAB2>cd Desktop

C:\Users\ITLAB2\Desktop>cd 25EU02088

C:\Users\ITLAB2\Desktop\25EU02088>javac P2.java

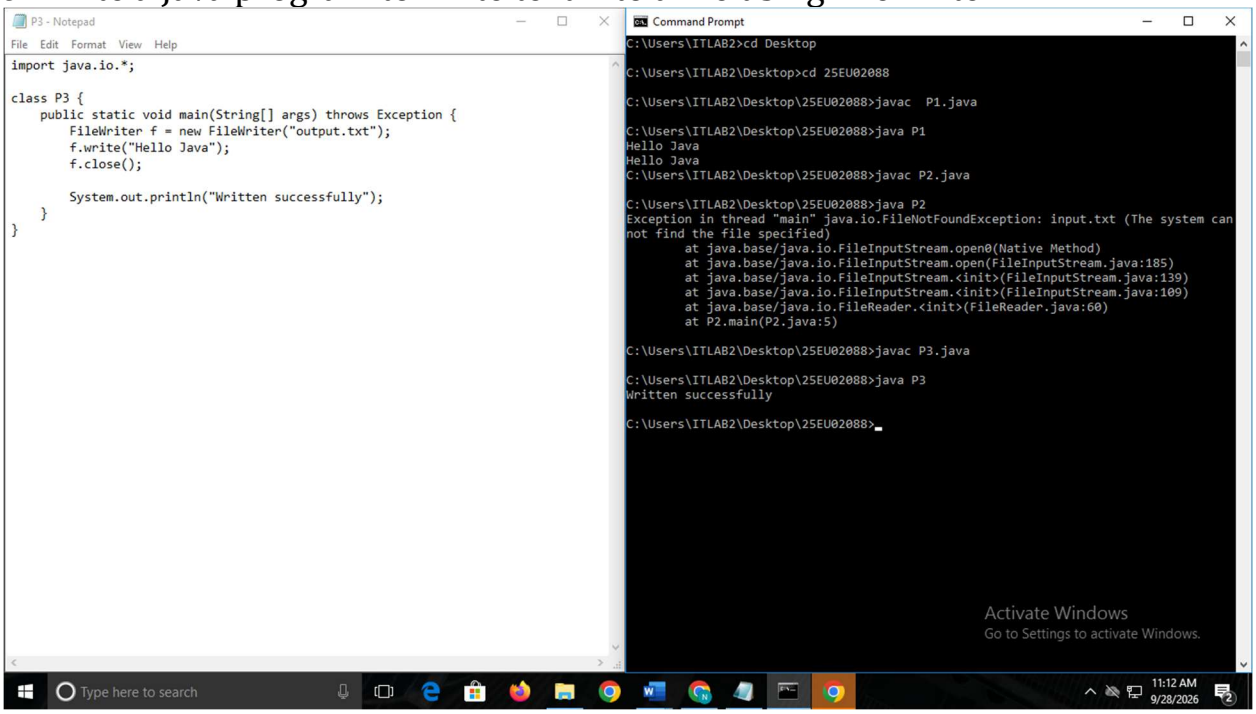
C:\Users\ITLAB2\Desktop\25EU02088>java P2
Hello java
C:\Users\ITLAB2\Desktop\25EU02088>
```

Activate Windows
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3. Write a Java program to write text into a file using FileWriter.



P3 - Notepad

File Edit Format View Help

```
import java.io.*;

class P3 {
    public static void main(String[] args) throws Exception {
        FileWriter f = new FileWriter("output.txt");
        f.write("Hello Java");
        f.close();

        System.out.println("Written successfully");
    }
}
```

Command Prompt

```
C:\Users\ITLAB2>cd Desktop

C:\Users\ITLAB2\Desktop>cd 25EU02088

C:\Users\ITLAB2\Desktop\25EU02088>javac  P1.java

C:\Users\ITLAB2\Desktop\25EU02088>java P1
Hello Java
Hello Java
C:\Users\ITLAB2\Desktop\25EU02088>javac P2.java

C:\Users\ITLAB2\Desktop\25EU02088>java P2
Exception in thread "main" java.io.FileNotFoundException: input.txt (The system can
not find the file specified)
        at java.base/java.io.FileInputStream.open0(Native Method)
        at java.base/java.io.FileInputStream.open(FileInputStream.java:185)
        at java.base/java.io.FileInputStream.<init>(FileInputStream.java:139)
        at java.base/java.io.FileInputStream.<init>(FileInputStream.java:109)
        at java.base/java.io.FileReader.<init>(FileReader.java:60)
        at P2.main(P2.java:5)

C:\Users\ITLAB2\Desktop\25EU02088>javac P3.java

C:\Users\ITLAB2\Desktop\25EU02088>java P3
Written successfully

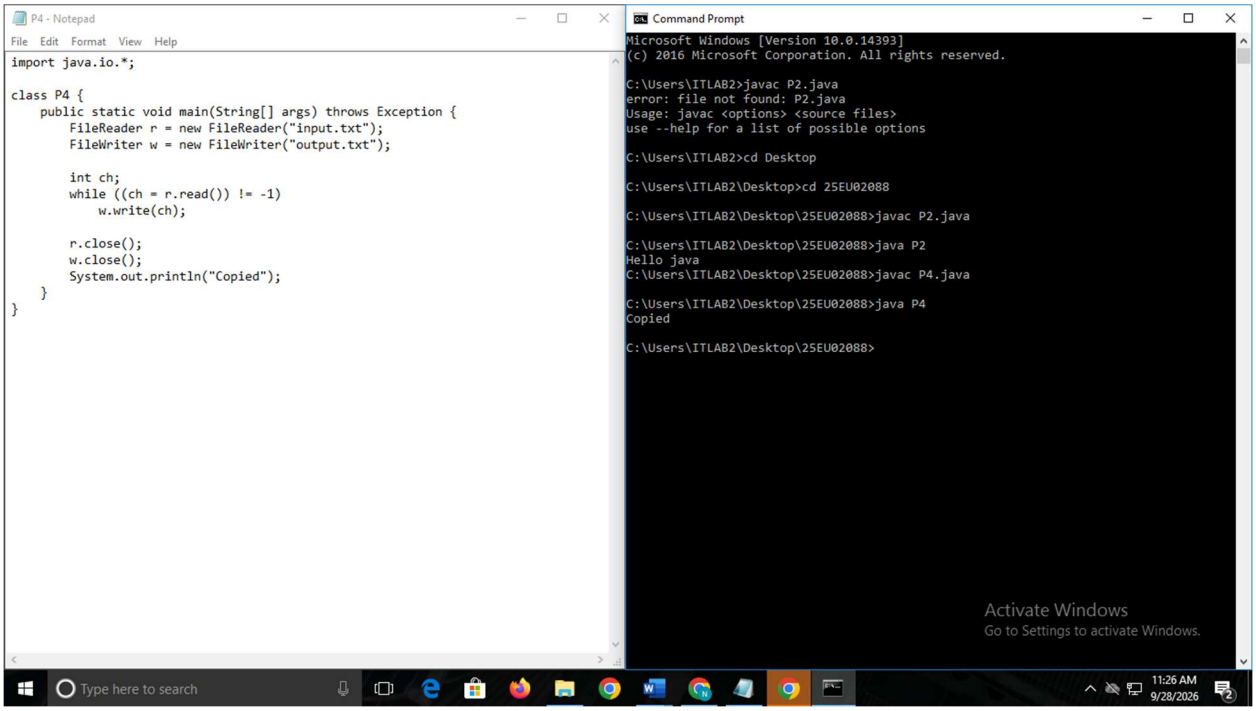
C:\Users\ITLAB2\Desktop\25EU02088>
```

Activate Windows
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9/28/2026

4. Write a Java program to copy the contents of one text file into another using FileReader and FileWriter.



P4 - Notepad

File Edit Format View Help

```
import java.io.*;

class P4 {
    public static void main(String[] args) throws Exception {
        FileReader r = new FileReader("input.txt");
        FileWriter w = new FileWriter("output.txt");

        int ch;
        while ((ch = r.read()) != -1)
            w.write(ch);

        r.close();
        w.close();
        System.out.println("Copied");
    }
}
```

Command Prompt

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2>javac P2.java
error: file not found: P2.java
Usage: javac <options> <source files>
use --help for a list of possible options

C:\Users\ITLAB2>cd Desktop

C:\Users\ITLAB2\Desktop>cd 25EU02088

C:\Users\ITLAB2\Desktop\25EU02088>javac P2.java

C:\Users\ITLAB2\Desktop\25EU02088>java P2
Hello java
C:\Users\ITLAB2\Desktop\25EU02088>javac P4.java

C:\Users\ITLAB2\Desktop\25EU02088>java P4
Copied

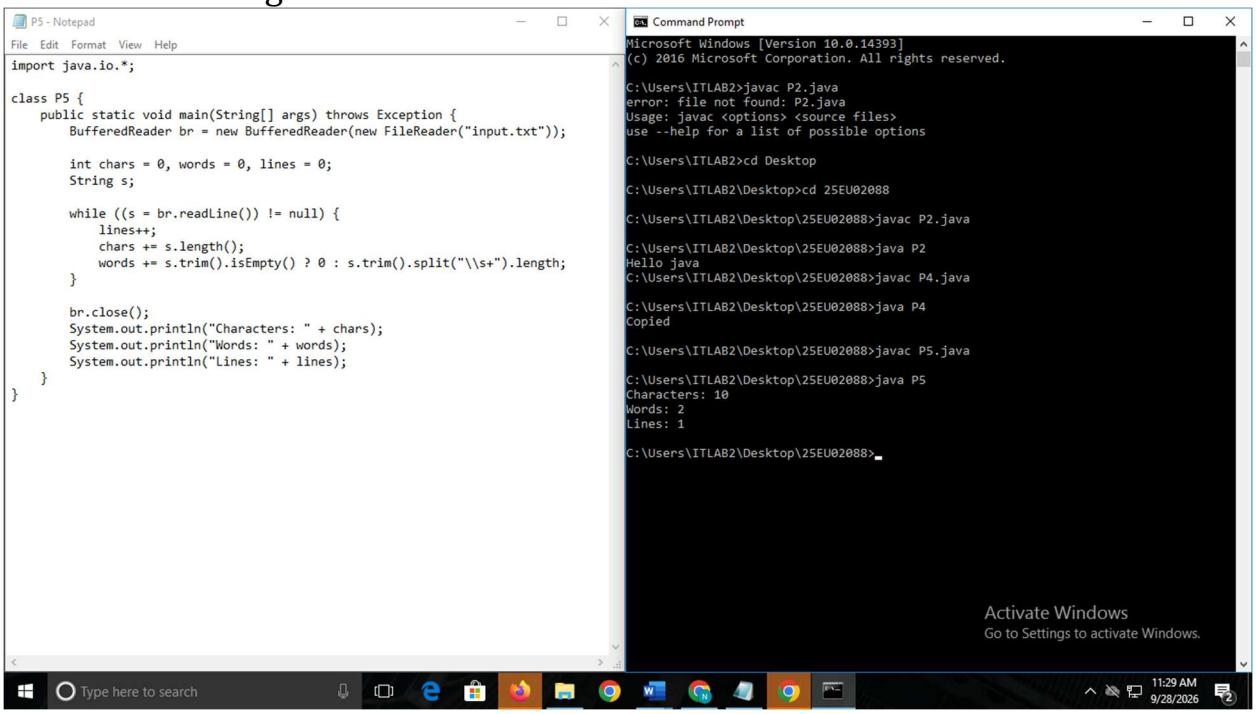
C:\Users\ITLAB2\Desktop\25EU02088>
```

Activate Windows
Go to Settings to activate Windows.

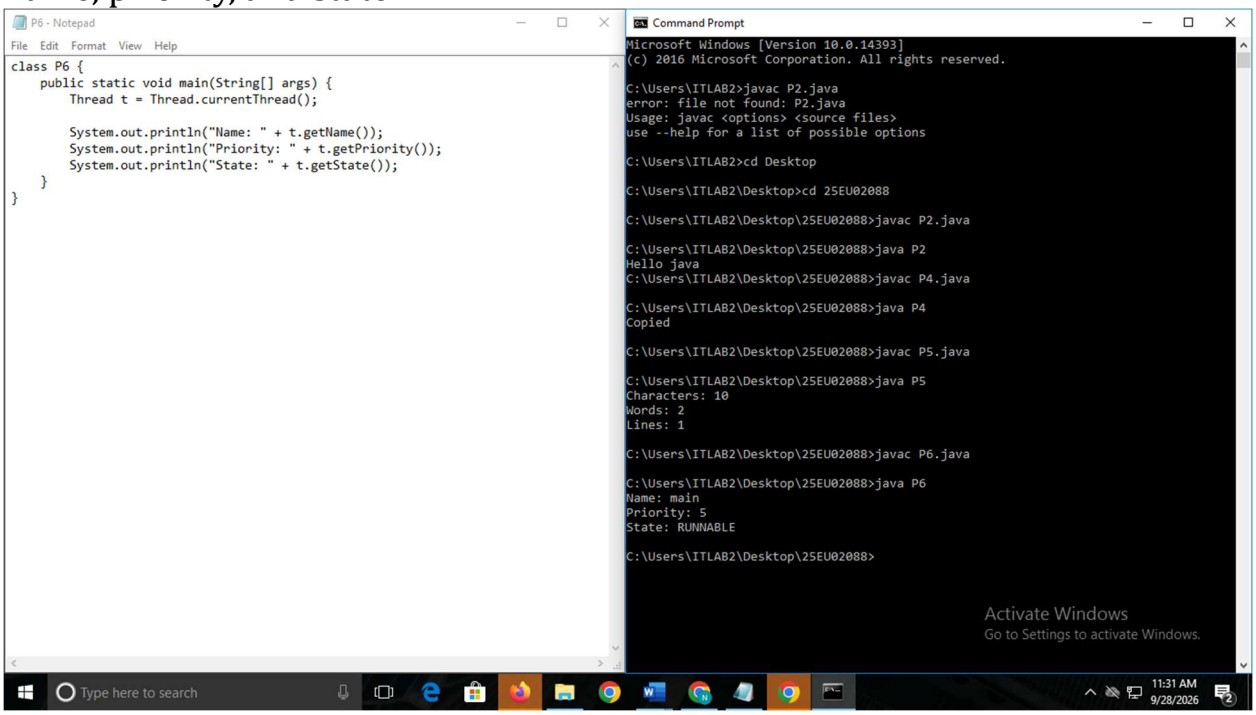
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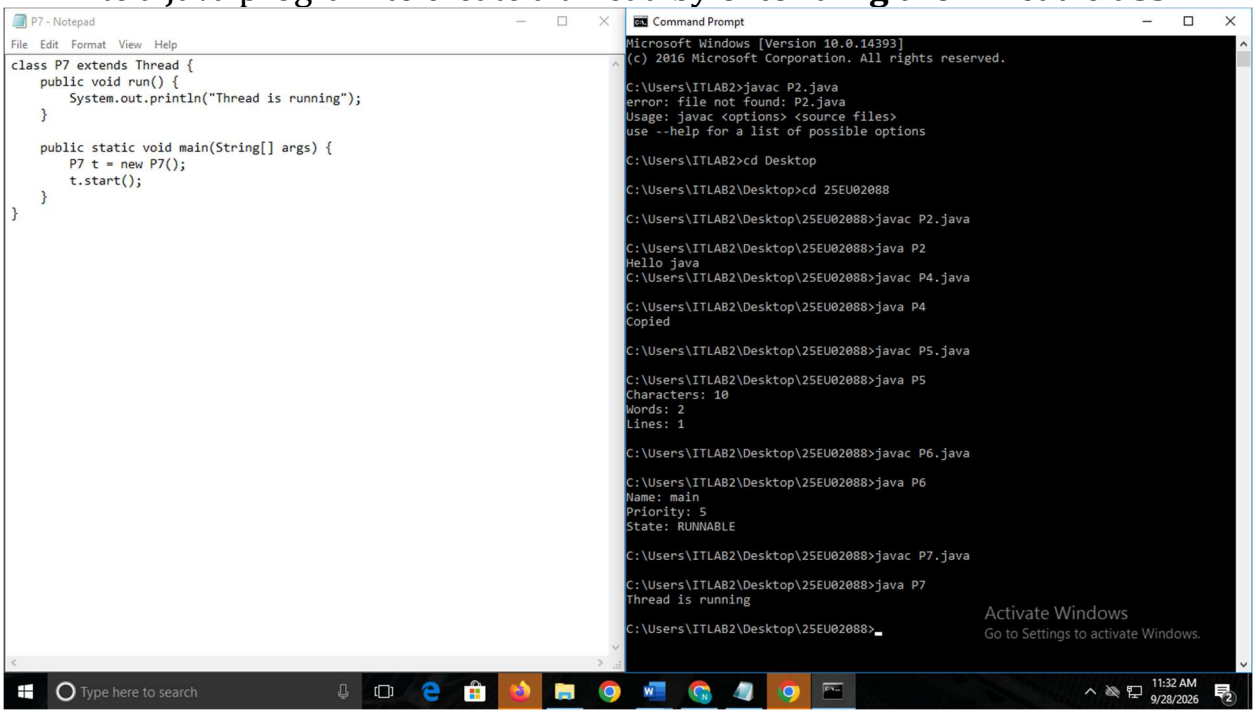
5. Write a Java program to count the number of **characters, words, and lines** in a text file using character streams.



6. Write a Java program to demonstrate the **Main Thread** and display its name, priority, and state.



7. Write a Java program to create a thread by **extending the** Thread **class**.



P7 - Notepad

File Edit Format View Help

```
class P7 extends Thread {
    public void run() {
        System.out.println("Thread is running");
    }

    public static void main(String[] args) {
        P7 t = new P7();
        t.start();
    }
}
```

Command Prompt

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2>javac P2.java
error: file not found: P2.java
Usage: javac <options> <source files>
use --help for a list of possible options

C:\Users\ITLAB2>cd Desktop

C:\Users\ITLAB2\Desktop>cd 25EU02088

C:\Users\ITLAB2\Desktop\25EU02088>javac P2.java

C:\Users\ITLAB2\Desktop\25EU02088>java P2
Hello java
C:\Users\ITLAB2\Desktop\25EU02088>javac P4.java

C:\Users\ITLAB2\Desktop\25EU02088>java P4
Copied

C:\Users\ITLAB2\Desktop\25EU02088>javac P5.java

C:\Users\ITLAB2\Desktop\25EU02088>java P5
Characters: 10
Words: 2
Lines: 1

C:\Users\ITLAB2\Desktop\25EU02088>javac P6.java

C:\Users\ITLAB2\Desktop\25EU02088>java P6
Name: main
Priority: 5
State: RUNNABLE

C:\Users\ITLAB2\Desktop\25EU02088>javac P7.java

C:\Users\ITLAB2\Desktop\25EU02088>java P7
Thread is running

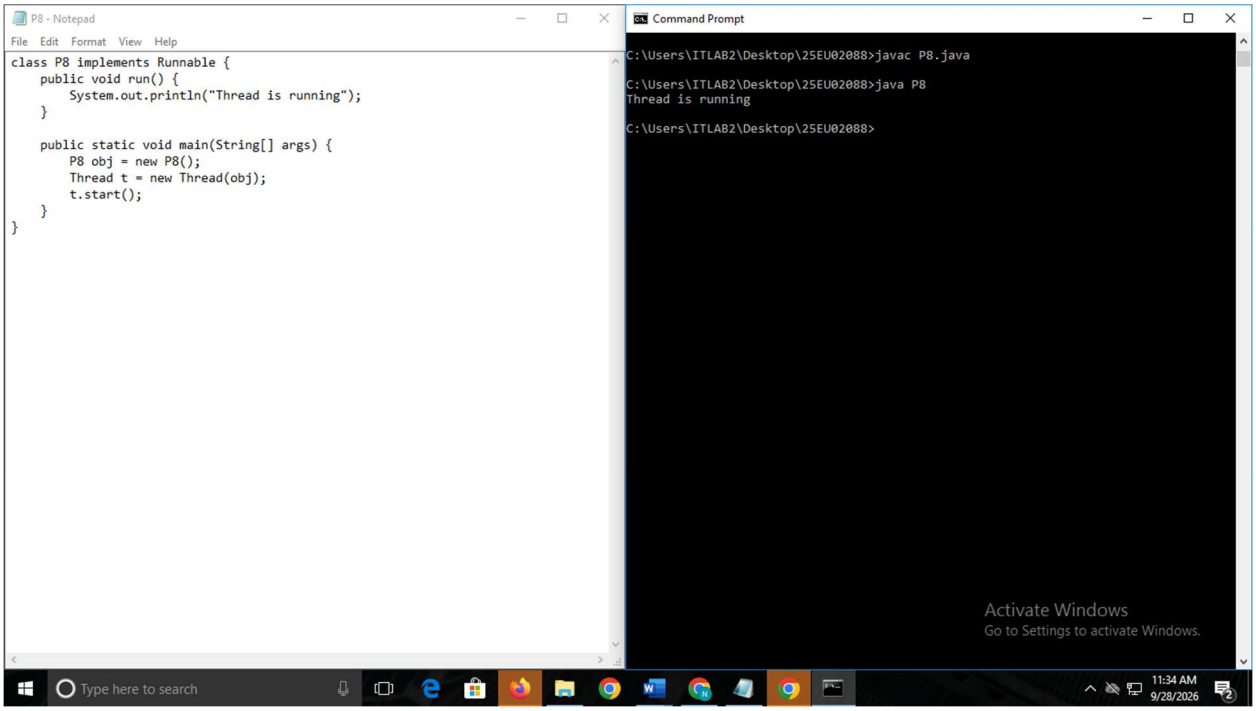
C:\Users\ITLAB2\Desktop\25EU02088>
```

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8. Write a Java program to create a thread by **implementing the** Runnable **interface**.



P8 - Notepad

File Edit Format View Help

```
class P8 implements Runnable {
    public void run() {
        System.out.println("Thread is running");
    }

    public static void main(String[] args) {
        P8 obj = new P8();
        Thread t = new Thread(obj);
        t.start();
    }
}
```

Command Prompt

```
C:\Users\ITLAB2\Desktop\25EU02088>javac P8.java

C:\Users\ITLAB2\Desktop\25EU02088>java P8
Thread is running

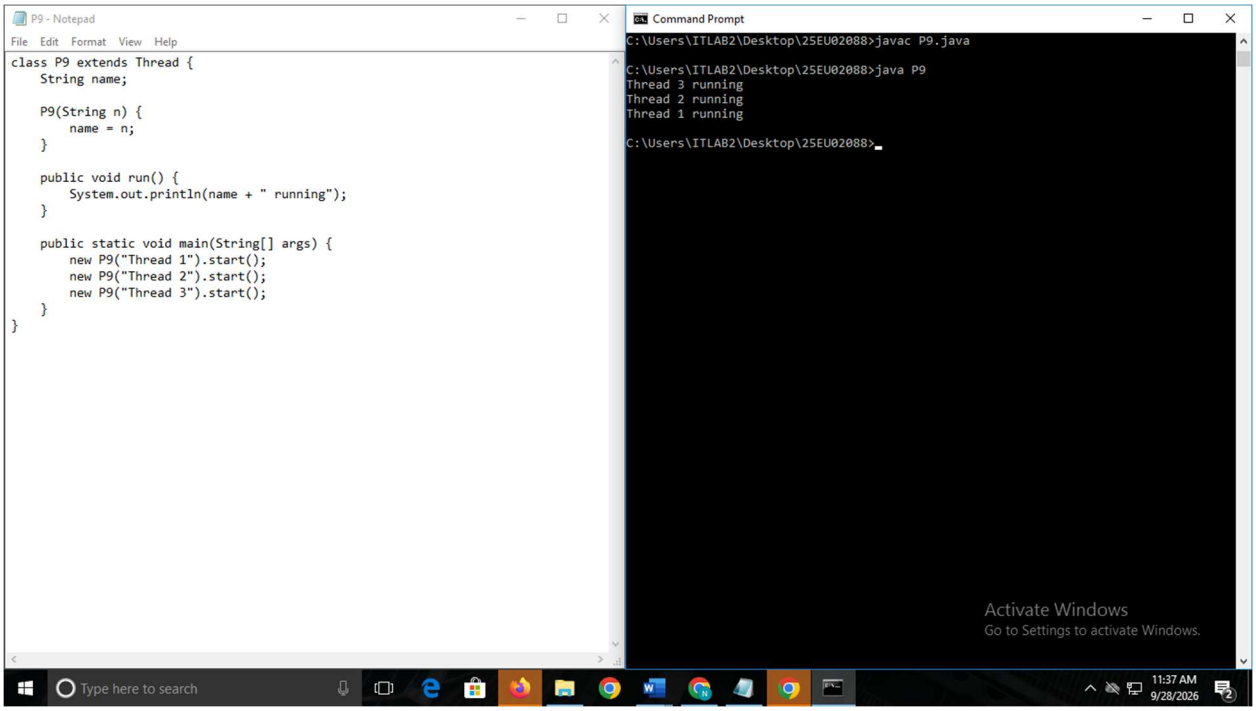
C:\Users\ITLAB2\Desktop\25EU02088>
```

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9. Write a Java program to create **multiple threads** and demonstrate their concurrent execution.



P9 - Notepad

File Edit Format View Help

```
class P9 extends Thread {
    String name;

    P9(String n) {
        name = n;
    }

    public void run() {
        System.out.println(name + " running");
    }

    public static void main(String[] args) {
        new P9("Thread 1").start();
        new P9("Thread 2").start();
        new P9("Thread 3").start();
    }
}
```

Command Prompt

```
C:\Users\ITLAB2\Desktop\25EU02088>javac P9.java

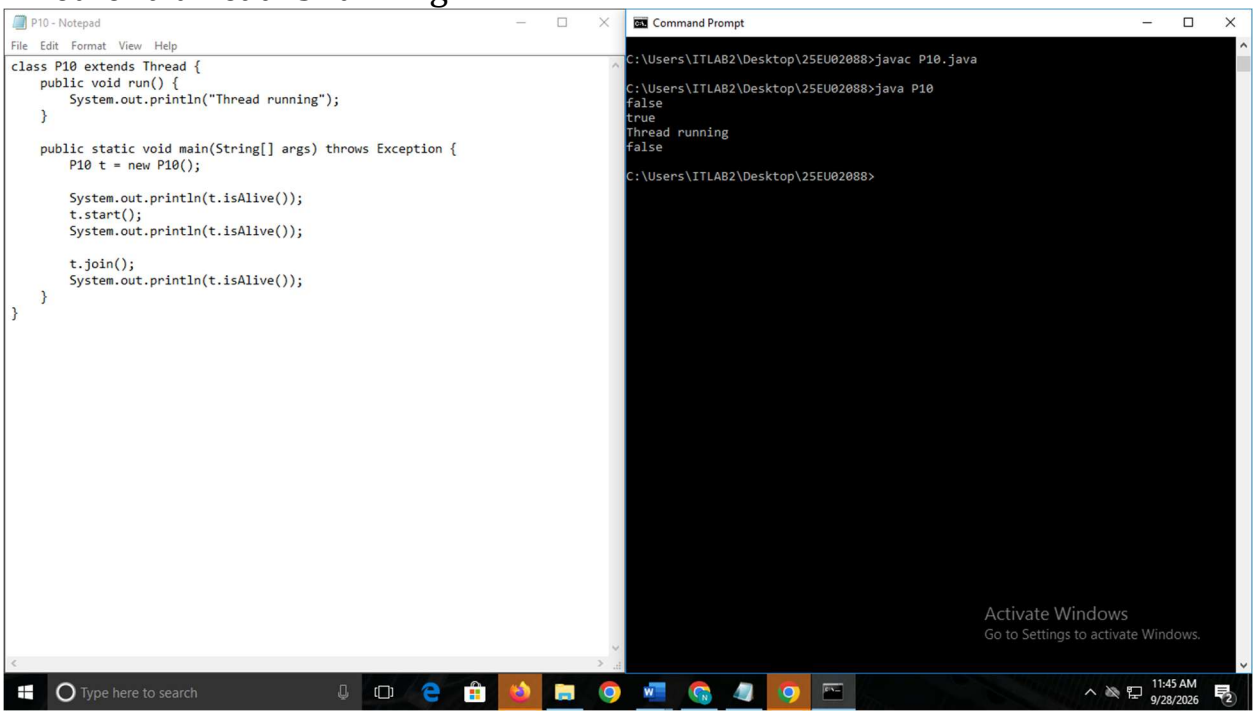
C:\Users\ITLAB2\Desktop\25EU02088>java P9
Thread 3 running
Thread 2 running
Thread 1 running

C:\Users\ITLAB2\Desktop\25EU02088>
```

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10. Write a Java program to demonstrate the use of isAlive() method to check whether a thread is running.



P10 - Notepad

File Edit Format View Help

```
class P10 extends Thread {
    public void run() {
        System.out.println("Thread running");
    }

    public static void main(String[] args) throws Exception {
        P10 t = new P10();

        System.out.println(t.isAlive());
        t.start();
        System.out.println(t.isAlive());

        t.join();
        System.out.println(t.isAlive());
    }
}
```

Command Prompt

```
C:\Users\ITLAB2\Desktop\25EU02088>javac P10.java

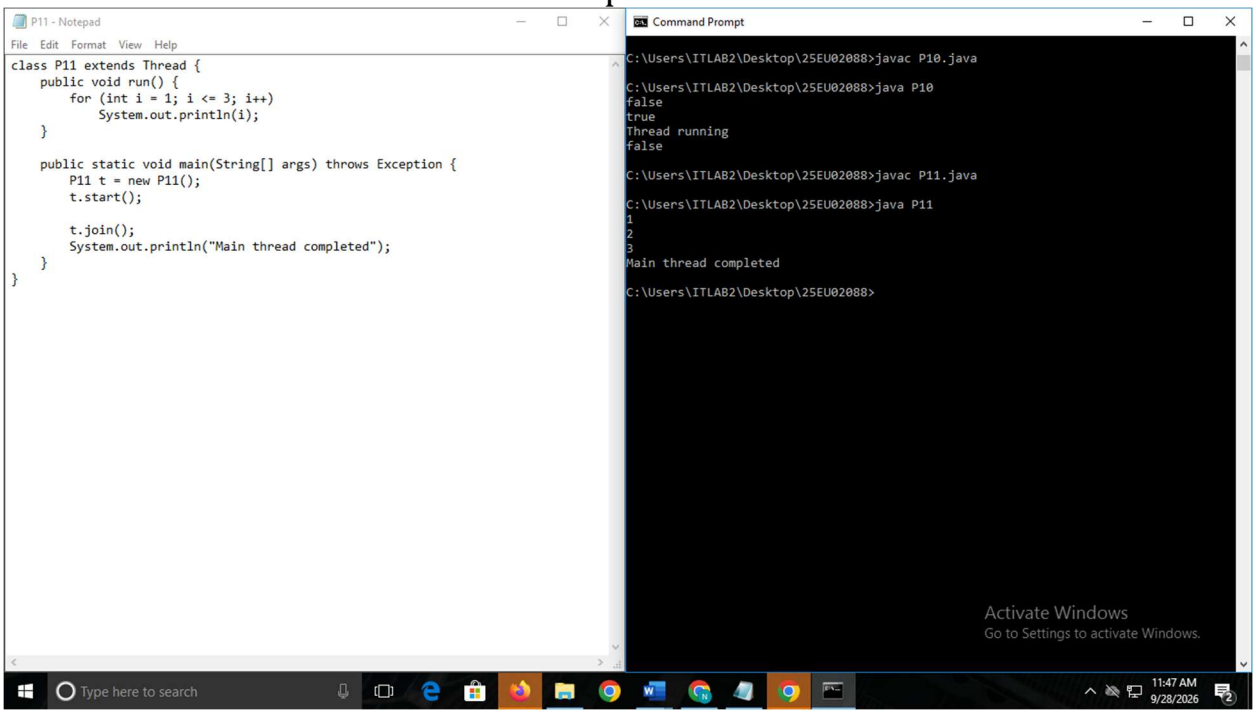
C:\Users\ITLAB2\Desktop\25EU02088>java P10
false
true
Thread running
false

C:\Users\ITLAB2\Desktop\25EU02088>
```

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11. Write Java program to demonstrate the use of join() method to make one thread wait for another thread to complete.



P11 - Notepad

File Edit Format View Help

```
class P11 extends Thread {
    public void run() {
        for (int i = 1; i <= 3; i++)
            System.out.println(i);
    }

    public static void main(String[] args) throws Exception {
        P11 t = new P11();
        t.start();

        t.join();
        System.out.println("Main thread completed");
    }
}
```

Command Prompt

```
C:\Users\ITLAB2\Desktop\25EU02088>javac P10.java

C:\Users\ITLAB2\Desktop\25EU02088>java P10
false
true
Thread running
false

C:\Users\ITLAB2\Desktop\25EU02088>javac P11.java

C:\Users\ITLAB2\Desktop\25EU02088>java P11
1
2
3
Main thread completed

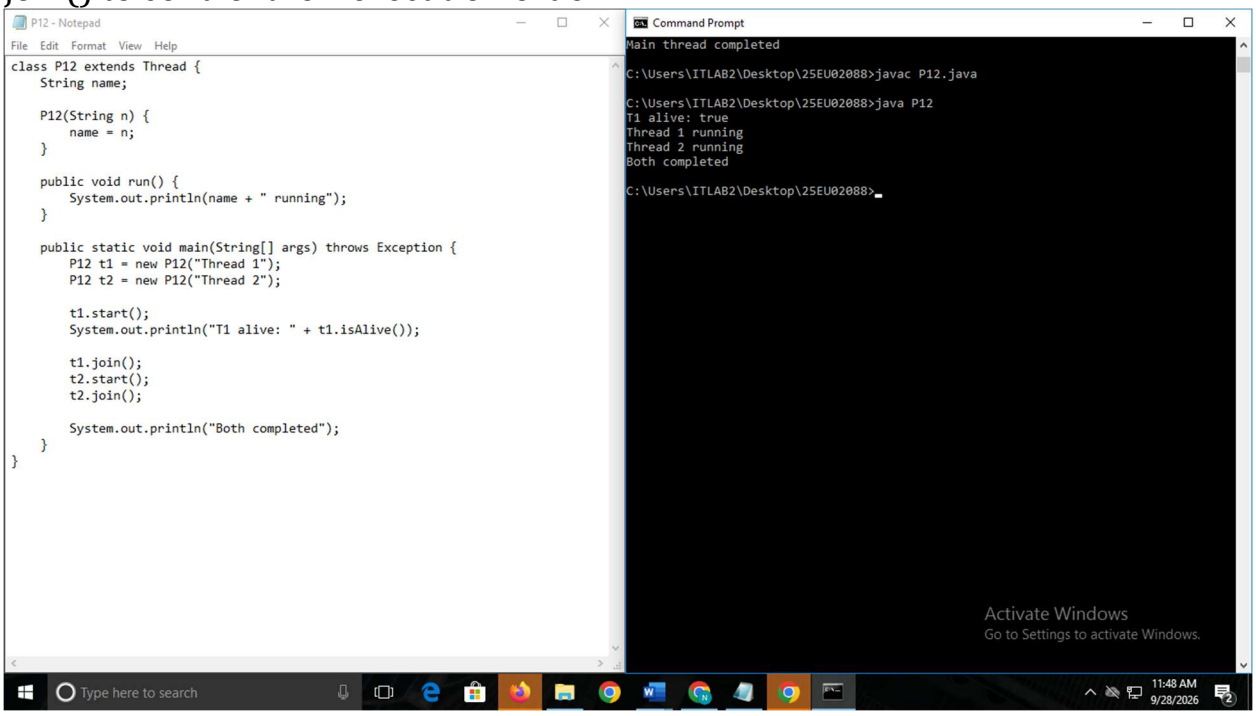
C:\Users\ITLAB2\Desktop\25EU02088>
```

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12. Write a Java program to create multiple threads and use isAlive() **and** join() to control their execution order.



P12 - Notepad

File Edit Format View Help

```
class P12 extends Thread {
    String name;

    P12(String n) {
        name = n;
    }

    public void run() {
        System.out.println(name + " running");
    }

    public static void main(String[] args) throws Exception {
        P12 t1 = new P12("Thread 1");
        P12 t2 = new P12("Thread 2");

        t1.start();
        System.out.println("T1 alive: " + t1.isAlive());

        t1.join();
        t2.start();
        t2.join();

        System.out.println("Both completed");
    }
}
```

Command Prompt

```
Main thread completed

C:\Users\ITLAB2\Desktop\25EU02088>javac P12.java

C:\Users\ITLAB2\Desktop\25EU02088>java P12
T1 alive: true
Thread 1 running
Thread 2 running
Both completed

C:\Users\ITLAB2\Desktop\25EU02088>
```

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